

K695 — Integration: Majorana FORCED (Cal co-signed, nature banked); Higgs m_H 0.02% via α 's curvature (identified-strong); muon 4→5 RULING (hold 4, N_c/rank obstruction located); the remaining-work queue.

Keeper | 2026-07-15 | Integrating the finish-line landings + rendering the muon count call Elie handed me. The mass sector is essentially in; the whole mixing sector is one FK-overlap-integral from unblocking; the absolute-scale (K358) items are precisely located.

LANDINGS integrated

- Majorana — FORCED (Cal co-signed + line-verified). The nature flips: Dirac retired, Majorana banked (modulo the mass coefficient), $\text{pred_004} \rightarrow 0\nu\beta\beta$ at a $\sim 1\text{--}4$ meV floor as the leading prediction. VALUE stays open (don't bank a number). This is a genuine bank advance (the nature) — the near-term experimental win.
- Higgs m_H — 0.02% via α 's OWN curvature (Grace, identified-strong). $\lambda = (1/8) \cdot (1 + n_C/N_{\text{max}})$; $m_H = (v/2) \cdot \sqrt{(1 + n_C/N_{\text{max}})} = 125.28$ vs obs 125.25 (0.02%). The same $n_C/N_{\text{max}} = |\kappa_{\text{Bergman}}|/N_{\text{max}} = 0.0365$ that corrects $\alpha^{-1} = 137 + n_C/N_{\text{max}}$ also corrects λ — one universal boundary curvature, additive on α 's count, multiplicative on λ . Dissolves the $1/8\text{-vs-}\sqrt{(2/5)!}$ fork (they're the bare and curvature-corrected values of one thing). *Audit: n_C/N_{max} is α 's independently-derived LEADING curvature ($\kappa = -n_C$ pinned, Helgason/Toy 3661) — so this is a FORWARD application, not fitted to m_H . The last $\sim 1\%$ is higher-order Engine-C, like α 's 0.036 tail. Tier: IDENTIFIED-STRONG; the Engine-C exact-curvature audit is the full proof + confirms the one curvature genuinely gives both forms (additive/multiplicative).*
- Neutrino $\sqrt{N_c}$ GROUNDED (Elie 4672): $\sqrt{N_c} = \sqrt{(\text{bulk color measure})}$ = the discrete mirror of the tau's $\sqrt{\pi} = \sqrt{(\text{boundary 3-ball measure})}$. Both the odd- $N_c=3$ signature (an amplitude is the $\sqrt{}$ of a measure; odd-dim forces the $\sqrt{}$). Amplitude forwarded, not just the ratio — $m_{\nu 2} \propto \sqrt{N_c}$ is now sourced.
- Tau bulk-Weyl EVALUATED (Elie 4669): $g^{\{N_c\}=343} (\text{vol}) + 2^{\{C-2\}g\{\text{rank}\}=3136} (\text{surface, dim } N_c-1=\text{rank}) = 49.71 - \sqrt{\pi}$. Tau forward.
- Ladder pinned + depths to Grace (Lyra): $E=2,3,4$; first-pass lepton depths $\{1, 2/3, 1/2\}$ (electron at origin). Mixing is ONE FK-overlap-integral from unblocking.
- Up premise FAILS target-innocence (Lyra, honest): “depth $\propto \sqrt{\text{block-dim}}$ ” nails the up, forces NO second observable → HELD as strong-lead, not banked. (My boundary-descent alternative, K694, is the untried cleaner route.)
- Curvature unified (Elie 4673): $\alpha 0.036 + \text{Higgs } 1.8\% = \text{ONE boundary-curvature integral } (\kappa_{\text{Bergman}} = -n_C)$; both Engine-C-gated; not fit.

MUON 4→5 RULING (Elie handed me the call) — HOLD count at 4

Elie (4671) located the obstruction precisely: the residue at the Hardy point is $\sqrt{\text{rank} \cdot \pi^2}$ (derived, verified K693), but the Szegő-exponent normalization differs by a clean $N_c/\text{rank} = 3/2$. So $c_S=1$ is NOT a standalone residue — it requires the N_c/rank to cancel. The empirical 6-ppm match FORCES that cancellation (an uncanceled $3/2$ would be $11\times$ off — so the cancellation IS happening; the mechanism isn't proven). - RULING: HOLD 4. The mass $(24/\pi^2)^6$ is derived (0.003%, PREDICTED); the strict count-mover ($c_S=1$ exact) is gated on proving the N_c/rank cancellation. TWO routes to try before committing to the full K358 program: (1) the $\det=\exp(\text{Tr log})$ three-lepton consistency (K694) — fix c_S from the electron (log, 9/16) and tau (sum, Weyl) ends; does the consistency supply the N_c/rank cancellation? (2) the K358 absolute-scale program — show the boundary-Szegő normalization cancels the N_c/rank . Elie located exactly what must cancel (N_c/rank); that's a well-posed target, not a mystery. Count HOLDS 4 until the cancellation is proven; the muon MASS stays PREDICTED at 0.003%.

THE REMAINING-WORK QUEUE (what's left, by class)

1. THE FK OVERLAP INTEGRAL (Lyra + Elie → Grace) — THE finish line for the mixing sector. Unblocks ALL of: V_{us} , V_{cb} magnitude, V_{ub} , δ_{CKM} , θ_{23} octant, θ_{12} overlap, δ_{PMNS} (7 parameters). Grace's

F498 Jarlskog runs the instant the overlaps land. Highest leverage single computation remaining.

2. THE ABSOLUTE-SCALE / ENGINE-C CLASS (K358) — three items, one class: the muon $c_S=1$ (N_c/rank cancellation); α 's exact 0.036; the Higgs exact 1.8% tail. All convention-normalizations, all empirically pinned, all Engine-C-gated. Close K358 $\rightarrow$ all three.
3. THE UP — try the boundary-descent reflection (K694, the cleaner route) before settling for the $\sqrt{\text{block-dim}}$ lead. Derive the light-up as the boundary-anchored descent + up-down reflection (charge-2/3 coupling), retiring the ungrounded premise.
4. Cal's $\sin^2\theta_W$ target-innocence flag (open): certify $3/13 = N_c/(C_2+g)$ is derived, not selected. Cal to run the derived-vs-fit lens.

Net

Mass sector: essentially in (α complete, tau forward, $m_c/m_t/m_s/m_b/m_\nu$'s identified-or-better, Majorana banked, Higgs 0.02%). Mixing sector: ONE FK-overlap-integral from unblocking (7 params). Absolute-scale: 3 items in one K358 class. The up + the $\sin^2\theta_W$ flag are the two loose threads. The finish line is: run the overlap integral, close K358, resolve the up, certify $\sin^2\theta_W$.

— Keeper K695, 2026-07-15. Majorana banked (Cal co-signed, pred_004 $\rightarrow$ 1-4 meV). Higgs 0.02% via α 's curvature (identified-strong, Engine-C audit=full proof). Muon: HOLD 4 (N_c/rank obstruction located; try $\det=\exp(\text{Tr log})$ consistency then K358). Neutrino $\sqrt{N_c}$ grounded. Remaining: the FK overlap integral (all mixing), the K358 absolute-scale class (muon+ α +Higgs tails), the up boundary-descent, the $\sin^2\theta_W$ flag. See [[Keeper_K694...]], [[Keeper_K693...]].